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|---|---------|--|----------------------|-----|
| 1 | (a) | average velocity = $540 / 30$
= 18 m s^{-1} | C1
A1 | [2] |
| | (b) | velocity zero at time $t = 0$
positive value and horizontal line for time $t = 5 \text{ s}$ to 35 s
line / curve through $v = 0$ at $t = 45 \text{ s}$ to negative velocity
negative horizontal line from 53 s with magnitude less than positive value and
horizontal line to time = 100 s | B1
B1
B1
B1 | [4] |
| 2 | (a) (i) | force is rate of change of momentum | B1 | [1] |